

Software Engineering for ‘Social Good’:

Integrating Action Research, Participatory Design, and Agile Development

Maria Angela Ferrario¹, Will Simm², Peter Newman², Stephen Forshaw³, Jon Whittle²

LUMS¹, School of Computing & Communications², LICA³, Lancaster University, UK

{m.a.ferrario; w.a.simm; p.newman; s.forshaw; j.n.whittle}@lancaster.ac.uk

ABSTRACT

Software engineering for ‘social good’ is an area receiving growing interest in recent years. Software is increasingly seen as a way to promote positive social change: this includes initiatives such as Code for America and events such as hackathons, which strive to build innovative software solutions with a social conscience. From a software engineering perspective, existing software processes do not always match the needs of these social software projects, which are primarily aimed at social change and often involve vulnerable communities. In this paper, we argue for new software processes that combine elements of agile, iterative development with principles drawn from action research and participatory design. The former allow social software projects to be built quickly with limited resources; the latter allow for a proper understanding of the social context and vulnerable user groups. The paper describes Speedplay, a software development management framework integrating these approaches, and illustrates its use in a real social innovation case study.

Categories and Subject Descriptors

D.2.9 [Management]: Software Process Models; K.2.4 [Social Issues]

General Terms

Management, Design, Human Factors.

Keywords

agile methods, action research, socially conscious software

1. INTRODUCTION

The last few years have seen a growing interest in software for ‘social good’, ranging from the development of on-line platforms for crowdsourcing support in disaster-stricken areas (e.g., Ushahidi), to ICT for Development (ICT4D) interventions in developing countries [25], and to initiatives targeting social needs in the developed world [16]. We argue that this social context presents the software engineering (SE) community with new opportunities as well as challenges in that it combines tight timelines and budget constraints typical of SE projects with ethical, governance and decision making issues [22] that are less typical of commercial SE environments.

In this paper, we define SE for ‘social good’ (henceforth, social

SE) to be the development, maintenance and sustainability of software aiming to promote social change. Our focus is on small-scale software that is developed for specific community needs where communities and software engineers work together.

Social SE’s primary motivation is social transformation. Here, software and digital artifacts are viewed as vehicles for social change, rather than end products or solutions to problems as in traditional commercial and non-commercial SE (e.g. Open Source Software). To this end, design and development decisions must be taken jointly with a diverse set of stakeholders including public service providers (local authorities, etc.), communities, charities and ‘hard to reach’ groups. Research [7] shows that this context requires the integration of SE methodologies, such as agile and iterative development (A-ID) [17], with user-focused and participatory approaches [8, 15, 21]. Crucially, such integration “*must exist within a cohesive project management framework that facilitates without being overly bureaucratic or prescriptive*” [7].

In this paper, we describe a novel project management framework, Speedplay, which does exactly this. Speedplay integrates A-ID development methods with approaches drawn from social science and design namely Action Research (AR) and Participatory Design (PD).

The definitions of AR and PD have been the subject of much debate. Here, we use the historic definitions of AR and PD. The father of AR, Kurt Lewin, defined AR in the 1940s as “*comparative research on the conditions and effects of various forms of social action*” that uses “*a spiral of steps, each of which is composed of a circle of planning, action, and fact-finding about the result of the action*” (as quoted in [14]). Sanders and Stappers [19] define PD as a democratic approach that views “*users as partners*” stressing the importance of the social dimension of work with technology. An extended abstract [10] previously introduced Speedplay from a social perspective; in this paper, we describe its SE process. The main research contribution of this paper is a novel software process integrating agile principles with elements of social science.

2. RELATED WORK

The value of innovative software in promoting social change is now well recognized. In recent years, there have been attempts to boost such efforts by using rapid prototyping to quickly develop applications for a social context. One such effort is the rise of hackerspaces, which are informal gatherings of software developers and representatives of community groups. These groups come together with the aim of developing software applications to address a social problem in a very short space of time (e.g., 24 hours or a weekend). Kera [16] followed and participated in a number of hackerspaces and found that they foster knowledge production and sharing between those involved, reinforcing a sense of community. Despite these tangible benefits, in many instances prototypes from hackathons do not have a life

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from Permissions@acm.org.

Copyright is held by the author/owner(s). Publication rights licensed to ACM.

ICSE Companion’14, May 31 – June 7, 2014, Hyderabad, India
ACM 978-1-4503-2768-8/14/05
<http://dx.doi.org/10.1145/2591062.2591121>

much beyond the initial event. Although many hackathons aim to address a particular social issue, developers' understanding of the issue is usually obtained informally, through dialogue with social representatives present at the hackathon. This ignores formal methodologies from social science that are aimed specifically at gaining a deeper understanding of social contexts. A key challenge when integrating SE and social science methods is that the latter typically demand many months, if not years, of in-depth study. This is in direct contradiction with agile SE methods, where the emphasis is on frequent delivery, ongoing evaluation and incremental development to closely match the user needs.

The integration of principles from UCD with SE has been explored as a possible method to strengthen collaboration between users and practitioners, especially in agile development. Like agile development, UCD has a core set of principles established in the 1980s [12]: *early focus on users and task, empirical measurement, and iterative design*. Sharp et al. contend that UCD and the customer collaboration seen in eXtreme Programming (XP) already share the aim of “*involv[ing] users effectively in software design*” [20]. Indeed, they state that the integration of UCD techniques may be *useful and welcomed* by the software community. This sentiment is echoed by Fox et al. who observed the successful practice of varying degrees of UCD applied to agile development in the software industry [11]. As previously noted, however, social SE main focus is social change; for this, we are interested in the integration of SE with PD and AR where the relationship between software developers and users can play equal parts in the transformation process.

An emerging challenge for social SE is how to sustain software once the initial development period, often linked to external funding, is over. The lack of focus on generating income means that the software will only be sustainable if volunteers devote their time to it. Siew and Yeo argue that tailoring the software to a specific local context is one way to achieve sustainability [21]. Localization can also help form a sense of community, which can help develop the community's capacity to “*innovate with and deploy ICT after the researchers withdraw*” [8].

To summarize, informal software processes such as hackathons face challenges related to software relevance and longevity. One way to address such issues is through a combination of HCI and SE methods. However, prior attempts to do so have been limited to combining SE with UCD. In UCD, the software developers still lead the process, and users tend to refine their ideas. Contextual Design, for example, “*deals with the front-end of design, from finding out who your customers are to testing a specific solution for them*” [3]. In contrast, we form *partnerships* with potential end-users from the start as we see software not as a solution, but as a means to collaboratively explore needs and aspirations for social transformation. This is aligned more with AR and PD than UCD.

3. SPEEDPLAY 4-STEP PROCESS MODEL

A key motivation behind the Speedplay process is to enable the development of software with an emphasis on social innovation in tightly constrained environments. Inspiration has been drawn from agile development, a framework of values and principles [2] such as flexibility, informal collaboration, and working code, that permit SE to be dynamic and engage end-users in steering the development process. These attributes help to enable software projects to “*survive in an atmosphere of constant change*”, and often results in customers quickly and frequently receiving requested functionality [6].

Speedplay in Context: speedplay has developed from our own extensive experience [23, 24] of working in partnership with stakeholders and end-users to tackle social issues within relatively short time frames (typically of 7-9 month duration) and with limited budgets. A-ID methods are often used in SE contexts to manage the development of software that requires stakeholder (owner) involvement for a rapid output delivery within tight budget and time constraints, hence they are appropriate for our development context. However, our projects work through community-academic partnerships [13] where stakeholders are not simply ‘involved’ in the software development process as in [6] but where they jointly set the research agenda and pro-actively engage in the design and development process. This context poses unprecedented challenges as well as opportunities to the SE community in at least three key areas: how the project constraints are managed; the way the stakeholders are engaged and their requirements captured in the design process; how the technology develops. Speedplay has emerged to address such challenges.

Speedplay as a Management Framework: from the theory perspective, Speedplay is an interdisciplinary project management framework that applies a streamlined version of the PRINCE2 management methodology [18] to a socially transformative research, collaborative design, and agile development process which embodies the principles of AR [14], PD [19] and A-ID [17]. We argue that, taken in isolation, the above methodologies are not suited for a software development context that aims to deliver a) working digital prototypes with social impact, b) within a short time frame and c) in partnership with a range of stakeholders that includes hard-to-reach groups. From the SE research perspective, this paper contributes to research looking at how SE methods can integrate with user-focused approaches including UCD [6], AR [21], and plan-driven methods [1]. The Speedplay process model in Figure 1 shows each step flowing into the next and how the process is *paced* through the delivery of tangible outputs or prototypes at each step. The prototypes may consist of both hardware and software and evolve from very early paper sketches to working prototypes. This output-driven, ‘disciplined’ [5] approach is inspired by PRINCE2 but it is different in that Speedplay produces minimal project management documentation, usually limited to monthly highlight reports. The three steps are of similar duration but of different ‘pace’ where the pace is relative to the time-lapse between each prototype version release. This is represented by the narrowing of the spiral from step 1 (slowest, a 10-12 week cycle) at the bottom, to step 3 (fastest, 2-3 week cycles) at the top.

Each step is characterized by a specific theoretic approach: the **prepare step** is grounded in AR principles and deploys qualitative research methods for initial user requirement capture. Key for ramping up the pace of this first step is the **focal point**, an event usually scheduled within the first 10 weeks of the project requiring the team to collaboratively produce a tangible output for an external audience. The **design step** embeds AR and PD principles into the design process and aims to visualize and design systems that can address user needs. This helps with the refinement of the requirements captured during step 1. The **build step** adopts a more traditional agile approach with short development cycles; it further refines the user requirements and concludes with the release of a stable technology prototype. Stakeholders and end-users are involved at every step of the process, particularly in the **sustain step** where wider partnerships are sought to support long-term development and deployments of the prototypes. The refinement of the user requirements is represented by the tapering of the spiral in the diagram (Figure 1).

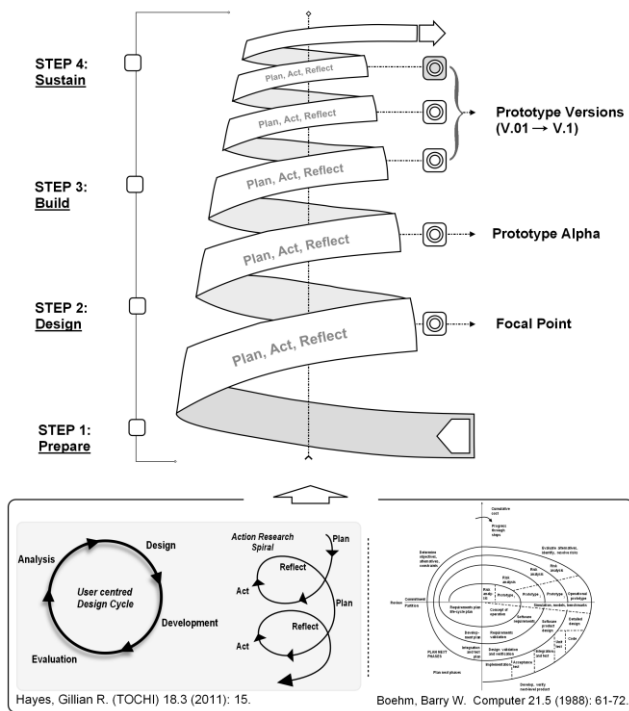


Figure 1: Speedplay Process Model

Figure 1 also includes three icons which serve two purposes: firstly, they highlight the difference between AR and UCD and how these differences relate to Speedplay; secondly, they compare and contrast Speedplay with traditional SE spiral models [4]. AR brings an *incremental* transformation of social practices through the continuous *iterations* of three phases: *plan*, *act* and *reflect*, hence the upward spiral; UCD instead supports a more traditional, albeit user-focused, *iterative* design development through *design*, *development*, *evaluation* and *analysis* cycles [14]. We argue that the Speedplay approach is, like AR, *incremental* and works through iterations of planning, action and reflection, however, it allows specific methods and techniques drawn from both PD and AR to be adopted in parallel, separately or in combination depending on the context. The Speedplay model shares a similarity with Boehm's spiral in that it is iterative; however, Boehm's is meant for large-scale projects whereas Speedplay emerged from small-scale projects. In addition, Boehm's spiral model is risk-driven, whereas Speedplay helps to *pace* uncertainties. In other words, Speedplay has an opportunistic approach to risk; in Speedplay, risks are treated as an uncertainty that has the potential to be harnessed, mitigated or discarded.

4. ACCESS ASD CASE STUDY

In this section, we describe a case study of Speedplay in practice in a research project. In the Access ASD (AASD) project, a partnership involving a multidisciplinary team of academics, care-service managers from a local authority, mental-health therapists and practitioners from national Autism charities came together for a nine-month research project. This partnership formed the research team, and each took an equal role in defining the research. The brief was intentionally wide: investigate technology applied to social anxiety experienced by adults diagnosed with High Functioning Autism/Asperger Syndrome (HFA/A). By adopting the Speedplay approach, a technical prototype called Clasp emerged from a deep insight into anxiety experienced by HFA/A adults gained through the Speedplay process. The latest

version of Clasp [23] is a novel tactile anxiety management, communication & peer support tool. Clasp connects a tactile anxiety coping device to a smartphone, which records and communicates anxiety levels for self-feedback and reflection. The data may be shared for real time online social network support, and aggregated for community feedback. Clasp¹ was prototyped with a Bluetooth connected stress ball, Android smartphone and the Diaspora social network.

Table 1 - Speedplay Applied to AASD

Step	Goal / Objectives	Output
1.Prepare	Goal: Build trust & up-skill - Understand epidemiology/theory of ASD - Identify HFA/A specific needs - Review ASD Digital tools - Initial idea generation with stakeholders - Initial users requirement capture	Design Brief
Focal Point (TTW5)	Goal: Ramp up pace and motivation - Brainstorm new ideas - Advance prototype design	Proof Of Concept
2.Design	Goal: Agree on a prototype system - Test tech feasibility and originality - Refine user requirements	Prototype α
3.Build	Goal: Build and test prototype system - Collaborative system development - Feedback from users/stakeholders - Finalise requirements - Future design implications	Prototype v1.0 (Final)
4.Sustain	Goal: Plan for future system development - Identify community/academic champions - Share system technical understanding - Collate and Share lessons learned	Follow-up bids

Table 1 shows the AASD goal, objectives and deliverables for each Speedplay step. Through the **prepare** step, the research team up-skilled each other in key knowledge areas, and researchers began to engage with a wide number of HFA/A adults. Training was given by each set of stakeholders to the others, and deep dive literature reviews conducted. In collaboration with HFA/A adults, the team explored a wide range of emergent potential directions. This process not only came up with ideas, but by exploring those ideas the manifestations and experiences of HFA/A adults emerged. A **focal point** event brought together the ideas and allowed reflection on the process. The focal point was the sixth Tiree Tech Wave, a meeting of software engineers and digital makers on the isle of Tiree² where the idea for a multi-modal stress-management tool emerged. During the **design** step, the group of engaged (HFA/A) participants was reduced and focussed to form a 10 person core user group (CUG). Through a cycle of workshops and public show and tells, the CUG shaped the design of Clasp. Feedback on initial prototypes was sought from social care professionals outside of the project team before entering the build phase. In the **build** phase, we followed agile principles for prototype development, with real users engaged in iterative development cycles and Scrum-like meetings. At each 2-week iteration, we delivered an increment of a functioning prototype to the CUG. Each iteration of Clasp was reviewed, evaluated, informed and reflected upon by the CUG. The **sustain** step began

¹ Clasp video <http://www.youtube.com/watch?v=WjvLZrEYE7M>

² <http://tireetechwave.org/>

during the last few months of AASD. The Clasp system was demonstrated and presented to a variety of partners with the intention of exposing the research outcomes, whilst actively looking for ‘champions’. Further funding bids from the engaged group of users, community organizations and academics to develop and evaluate Clasp are under-review at the time of publication, and the code and schematics are to be released as an open-source project with the intention of benefitting a wide population, including beyond HFA/A.

5. CONCLUSIONS

This paper describes Speedplay, a novel project management framework that integrates A-ID development with methods drawn from Action Research (AR) and Participatory Design (PD). There are six fundamental aspects that describe its style that we argue are key to social software development: 1) Speedplay works in partnership with all stakeholders, including end-users; 2) it supports team building by engaging team members in cross-cutting tasks (e.g. the software developer engages in ethnographic observations whilst the designer familiarizes with the code); 3) it promotes self-direction by matching primary responsibilities to individual skills; 4) it applies, where appropriate, a variety of social research methods from qualitative to quantitative to capture and refine user requirements; 5) it systematically uses design and problem solving techniques to promote both critical and lateral thinking; 6) it delivers digital technology prototypes that are ready for further roll-outs by adopting A-ID approaches. Speedplay emerged from a number of social innovation projects, each up to 9 months in duration. It has been evaluated on three projects to date. One of these projects is on-going; in the other two cases, the stakeholders continue to develop the prototypes independently of the original funded project.

6. ACKNOWLEDGEMENTS

We thank Autism Initiatives, Lancashire County Council Adult Social Care Services, the NHS and especially the volunteers who took part in the Access ASD project CUG. This work was funded by RCUK Grant EP/I033017/1.

7. REFERENCES

- [1] Al-Zoabi, Z. (2008). Introducing Discipline to XP: Applying PRINCE2 on XP Projects. *3rd International Conference on Information & Communication Technologies: From Theory to Applications*. Apr. 2008, 1–7.
- [2] Beck, K. et al. (2001). Manifesto for Agile Software Development. <http://www.agilemanifesto.org>
- [3] Beyer, H., & Holtzblatt, K. (1999). Contextual design. *Interactions*, 6(1), 32–42.
- [4] Boehm, B. W. (1988). A spiral model of software development and enhancement. *Computer*, 21(5), 61–72.
- [5] Boehm, B., & Turner, R. (2004, May). Balancing agility and discipline: Evaluating and integrating agile and plan-driven methods. *26th International Conference on Software Engineering*, 718–719.
- [6] Capiluppi, A. et al. (2007). An Empirical Study of the Evolution of an Agile-Developed Software System. *29th International Conference on Software Engineering (ICSE '07)*. May 2007, 511–518.
- [7] Chamberlain, S., Sharp, H., & Maiden, N. (2006). Towards a framework for integrating agile development and user-centred design. In *Extreme Programming and Agile Processes in Software Engineering*, 143–153.
- [8] Dearden, A. and Rizvi, H. (2008) Adapting Participatory and Agile Software Methods to Participatory Rural Development. *10th Participatory Design Conference*, 224–228.
- [9] Dybå, T. and Dingsøyr, T. (2008). Empirical studies of agile software development: A systematic review. *Information and Software Technology*. 50 (9–10), 833–859.
- [10] Ferrario, M. A., Simm, W., & Whittle, J. (2013). Speedplay, Managing the Other Edge of Innovation in the Digital Economy. *Digital Futures*.
- [11] Fox, D. Sillito, J. & Maurer, F. (2008) Agile Methods and User-Centered Design: How These Two Methodologies are Being Successfully Integrated in Industry. *Agile Conference 2008*, 63–72.
- [12] Gould, J. & Lewis, C. (1985) Designing for Usability: Key Principles and What Designers Think. *Communications of the ACM*, 28(3) 300–311.
- [13] Hart, A., & Wolff, D. (2006) Developing local communities of practice through local community–university partnerships. *Planning, Practice & Research*, 21(1), 121–138.
- [14] Hayes, G. R. (2011) The relationship of action research to human-computer interaction. *ACM Transactions on Computer-Human Interaction (TOCHI)*, 18(3), 15.
- [15] Kautz, K. (2010) Participatory Design Activities and Agile Software Development. In *Human Benefit through the Diffusion of Information Systems Design Science Research* 24(3), 217–235.
- [16] Kera, D. (2012) Hackerspaces and DIYbio in Asia: connecting science and community with open data, kits and protocols. *Journal of Peer Production*, 1–8.
- [17] Larman, C. (2004) Agile and iterative development: a manager's guide. Addison-Wesley Professional.
- [18] Murray, A. (2009) *Managing successful projects with PRINCE2*. Stationery Office Books.
- [19] Sanders, E. B-N., Stappers, P. J. (2008) Co-creation and the new landscapes of design. *Co-Design* 4(1), 5–18.
- [20] Sharp, H., Robinson, H. and Segal, J. (2009) Integrating User-Centered Design and Software Engineering: A Role for Extreme Programming? Presentation available from <http://www.ics.heacademy.ac.uk/>
- [21] Siew, S. and Yeo, A.W. (2008) Employing Participatory Action Research to Augment Software Development for Rural Communities. *25th BCS Conference on Human-Computer Interaction*, 171–176.
- [22] Sigfridsson, A., Avram, G., Sheehan, A., Sullivan, D. (2007) Sprint-driven development: working, learning and the process of enculturation in the PyPy community. *Int. Fed. for Inf. Processing Digital Library*, 234(1) 133–146.
- [23] Simm, W., Ferrario, M. A., Southern, J., & Whittle, J. (2013) Clasp: Digital Tactile ASD Anxiety Management. *Digital Futures*.
- [24] Southern, J., Ellis, R., Ferrario, M. A., McNally, R., Dillon, R., Simm, W., & Whittle, J. (2013). Imaginative labour and relationships of care: Co-designing prototypes with vulnerable communities. *Technological Forecasting and Social Change*. In Press. Available online.
- [25] Tarafdar, M., Singh, R., & Anekal, P. (2013). Impact of ICT-enabled product and process innovations at the Bottom of the Pyramid: a market separations perspective. *Journal of Information Technology*. In Press. Available online.
- [26] Whittle, J., Simm, W., Ferrario, M. A., et al. (2010). VoiceYourView: collecting real-time feedback on the design of public spaces. *12th ACM International Conference on Ubiquitous Computing*, 41–50.